

Correct Answers, Uneven Grounding: The Distributed-Evidence Gap in Legal RAG

An evidence-shape analysis across a 100-question legal benchmark and 590 PDF pages

Answers stay level.

Grounding separates when evidence spans pages.

01 HYPOTHESIS

RAG can find enough evidence to answer while still emitting an incomplete citation set [1,3]. In legal QA, one correct page is not the same as complete support.

RESEARCH QUESTION

How does distributed evidence affect the relationship between answer quality and source grounding?

H1 Multi-page evidence will leave answer scores comparable but lower final page grounding; a uniform citation-narrowing policy may widen the gap.

95 / 100

questions had annotated gold pages and were eligible for this analysis

● 59 localized - one gold page

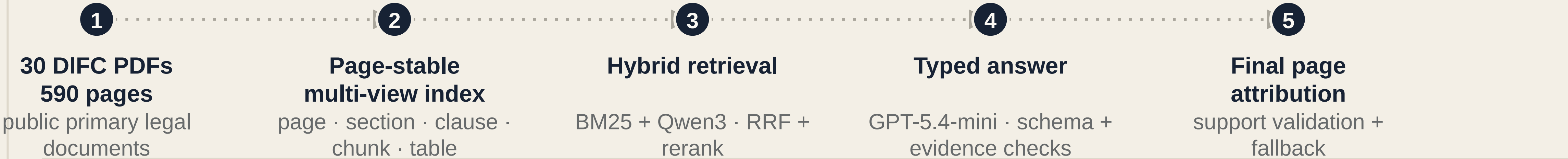
◆ 36 distributed - multiple gold pages

PAGE GROUNDING

Per-question $F\beta$ over emitted vs gold (document, page) pairs; $\beta = 2.5$ weights recall more than precision.

02 METHODOLOGY - PAGE-STABLE EVIDENCE FLOW

INVARIANT Every evidence record retains document_id + page_number.



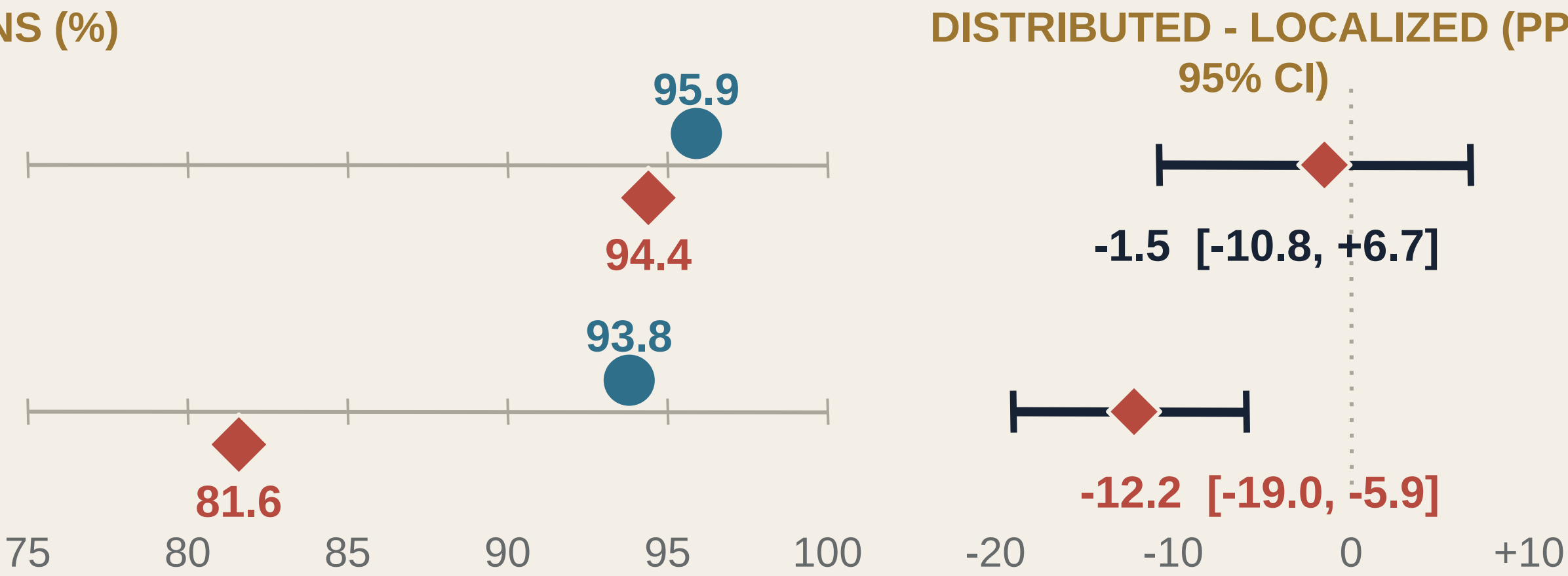
03 RESULTS

Answer quality stays close; final page grounding does not.

GROUP MEANS (%)

Answer quality

Final page grounding



PERFECT ANSWER, BUT PAGE GROUNDING < 0.8

Localized

Distributed

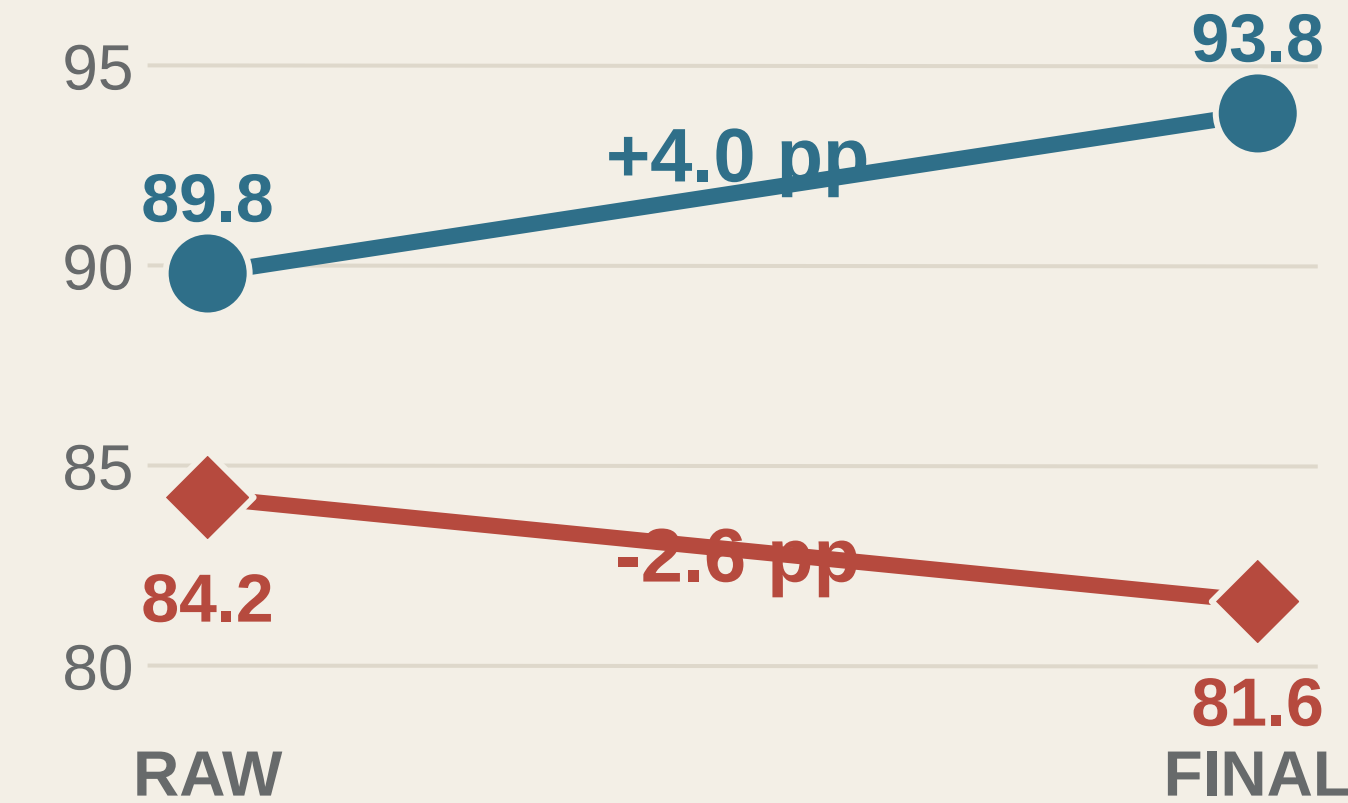
4 / 55 (7.3%)

16 / 34 (47.1%)

Denominators are perfectly scored answers, not all questions.

04 WHERE THE GAP WIDENS

Raw retrieval -> final attribution



DESIGN IMPLICATION

Treat evidence shape as a routing signal.

Narrow localized support; preserve and aggregate distributed support. At inference time, route on observable evidence spread or claim-level support. Observed association; not a causal ablation.

05 LIMITATIONS

- Single saved run; retrospective, non-preregistered hypothesis.
- Observational slice: page and document distribution are confounded (35/36 multi-page questions are also multi-document).
- Heterogeneous answer scoring and uneven answer-type composition.
- Raw-to-final comparison is diagnostic, not a causal ablation.

06 REPRODUCIBILITY

github.com/SteinVR/Agentic-RAG-Challenge
Saved artifacts: config, manifest, answer-score checkpoint, page ledger.
20,000 stratified bootstrap resamples; seed 20260903.
Corpus: DIFC Legal Database + DIFC Courts.

SELECTED REFERENCES

- [1] Lewis et al., NeurIPS 2020.
- [2] Cormack et al., SIGIR 2009.
- [3] Gao et al., EMNLP 2023.
- [4] Pipitone & Houir Alami, 2024.